



Mint Classics Inventory Analysis

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Overview

We are given the role of an entry-level data analyst at the fictional company called Mint Classics. Mint Classics is a retailer of classic model cars and other vehicles. The company is seeking suggestions for organizing and/or reducing inventory, while maintaining service to their customers. Our goal is to help to analyze data in a relational database with the goal of supporting inventory-related business decisions that lead to the closure to the closure of a storage facility at Mint Classics.





Questions for Guidance

1

Where are items stored and if they were rearranged, could a warehouse be eliminated?

2

Are inventory levels aligned with sales demand?

3

Are we storing items that are not moving?

Database Overview

Entity Relationship Diagram:

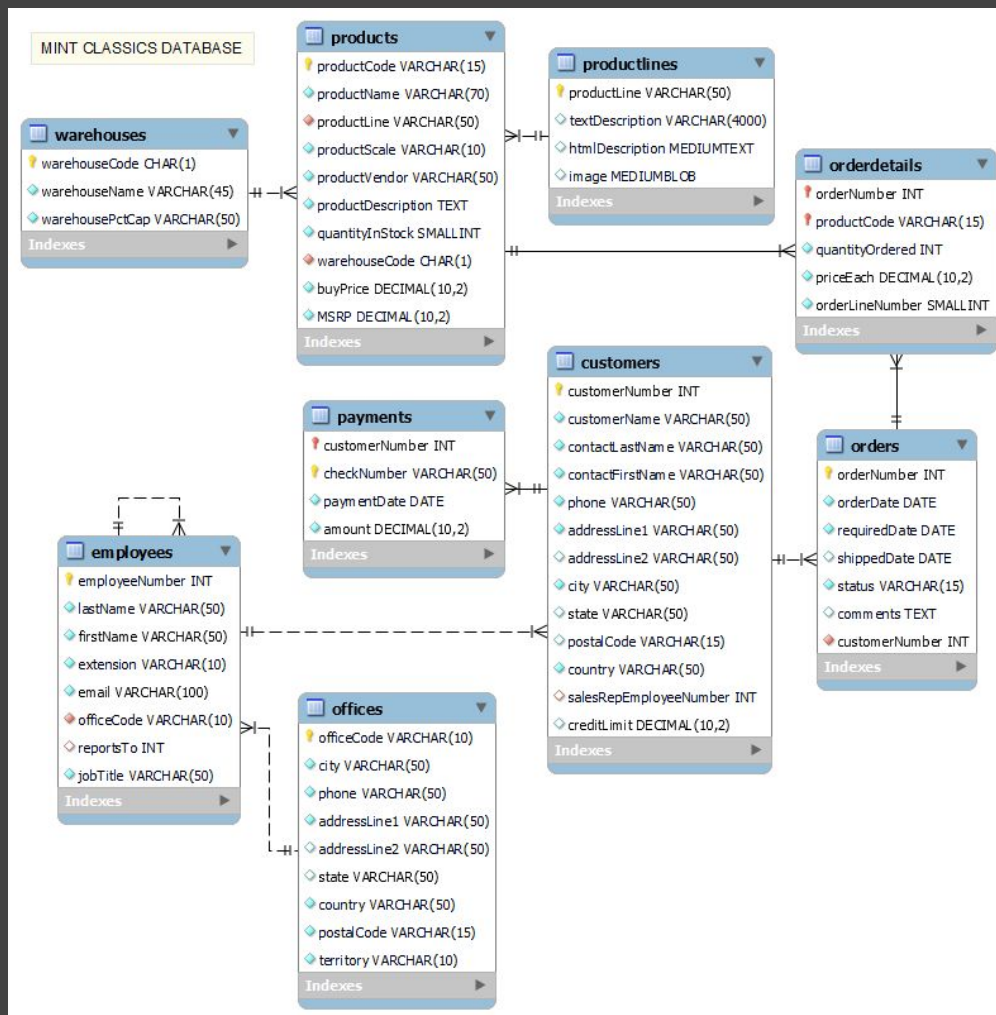
Main Tables Analyzed:

products - shows inventory, warehouse location, and identifying information for a product.

warehouses - shows capacity information for a given warehouse

orderdetails - shows product sales and price information

orders - shows customer order information



Warehouse Performance Overview

Warehouse	Capacity	Inventory Value	Units Sold	Revenue	Turnover
A	72%	\$6,660,000.00	24650.00	\$2,080,000.00	0.67%
B	67%	\$14,060,000.00	35582.00	\$3,850,000.00	0.61%
C	50%	\$5,700,000.00	22933.00	\$1,800,000.00	0.67%
D	75%	\$4,110,000.00	22351.00	\$1,880,000.00	1.02%

*Information gathered using MySQL from the Mint Classics Database



Key Observations

1: Warehouse B is the most important facility

Given that Warehouse B has the highest revenue, the highest sales volume, the largest inventory value, and the largest product variety, it would not be reasonable from a business perspective to close down this facility. Although Warehouse B has the lowest inventory turnover (0.61%), this may be due to the fact that it holds more inventory than the other warehouses

2: Warehouse C is a strong closure candidate

Given that Warehouse C contributes the least amount of revenue than any other facility while operating with half of its facility unused, Warehouse C is a strong closure candidate. Furthermore, it has sold the second-to-least amount of products. Its inventory value is also the second-to-least.



Slow-Moving Inventory Exits

SQL queries were ran to find products that were taking up warehouse space despite limited customer demand. From this output table from a SQL query, we can see that most of the slow-moving products are stored in warehouses B and C. We can make sense of the fact that warehouse B having slow-moving products given its large inventory, but warehouse C does not have as much inventory as warehouse B, and warehouse C is not utilizing its full capacity for storage of inventory.

	warehouseCode	productCo...	productName	quantityInSto...	total_sold
	b	S18_3233	1985 Toyota Supra	7733	NULL
	b	S18_4933	1957 Ford Thunderbird	3209	767
	b	S24_1046	1970 Chevy Chevelle SS 454	1005	803
	c	S24_3969	1936 Mercedes Benz 500k Roadster	2081	824
	c	S18_2248	1911 Ford Town Car	540	832
	b	S18_2870	1999 Indy 500 Monte Carlo SS	8164	855
	c	S18_4409	1932 Alfa Romeo 8C2300 Spider Sport	6553	866
	b	S24_4048	1992 Porsche Cayenne Turbo Silver	6582	867
	b	S24_3191	1969 Chevrolet Camaro Z28	4695	870
	b	S24_2887	1952 Citroen-15CV	1452	873
	c	S18_2795	1928 Mercedes-Benz SSK	548	880
	c	S18_3140	1903 Ford Model A	3913	883
	c	S24_3420	1937 Horch 930V Limousine	2902	884
	b	S24_3432	2002 Chevy Corvette	9446	894
	a	S700_1691	American Airlines: B767-300	5841	894
	d	S700_3962	The Queen Mary	5088	896
	d	S700_2047	HMS Bounty	3501	897
	a	S32_4485	1974 Ducati 350 Mk3 Desmo	3341	898
	d	S700_1938	The Mayflower	737	898
	b	S12_3990	1970 Plymouth Hemi Cuda	5663	900
	a	S32_2206	1982 Ducati 996 R	9241	906
	c	S18_3136	18th Century Vintage Horse Carriage	5992	907
	d	S32_1268	1980's GM Manhattan Express	5099	911
	b	S24_2972	1982 Lamborghini Diablo	7723	912
	b	S18_1589	1965 Aston Martin DB5	9042	914
	b	S18_3482	1976 Ford Gran Torino	9127	915
	b	S24_1628	1966 Shelby Cobra 427 S/C	8197	915
	b	S18_1984	1995 Honda Civic	9772	917

Final Recommendation:
Close Warehouse C!

We can arrive the conclusion that **Warehouse C should be closed** because the warehouse has the lowest revenue contribution to Mint Classics, it has the lowest capacity utilization (50%), it has a modest inventory value that can be potentially relocated to the other facilities, and closing Warehouse C will likely have a lower operational impact compared to closing any other warehouse.

Our other potential candidate for closure, **Warehouse B** (lowest inventory turnover), **should remain open** because it holds the highest revenue, the highest sales volume, the largest product variety, and is a critical contributor to customer fulfillment for Mint Classics.





Appendix

SQL queries

Query 1:

```
USE mintclassics;
SELECT warehouseCode, count(productCode) AS
number_of_items, SUM(quantityInStock) AS
quantity_in_stock
FROM products
GROUP BY warehouseCode;
```

Query 2:

```
USE mintclassics;
SELECT warehouseCode, warehousePctCap
FROM warehouses;
```

Query 3:

```
USE mintclassics;
SELECT p.productCode, p.productName,
p.quantityInStock, p.productDescription,
SUM(o.quantityOrdered) AS total_cars_sold
FROM products AS p LEFT JOIN orderdetails AS o
ON p.productCode = o.productCode
GROUP BY p.productCode
ORDER BY total_cars_sold DESC;
```



Appendix

SQL queries cont.

Query 4:

```
USE mintclassics;  
SELECT productCode, productName,  
quantityInStock, buyPrice, quantityInStock *  
buyPrice AS inventory_value  
FROM products  
ORDER BY inventory_value DESC;
```

Query 5:

```
USE mintclassics;  
SELECT w.warehouseCode, w.warehouseName,  
SUM(p.quantityInStock * p.buyPrice) AS  
inventory_value  
FROM warehouses AS w INNER JOIN products AS  
p  
ON w.warehouseCode = p.warehouseCode  
GROUP BY w.warehouseCode, w.warehouseName  
ORDER BY inventory_value DESC;
```



Appendix

SQL queries cont.

Query 6:

```
USE mintclassics;
SELECT p.warehouseCode,
SUM(o.quantityOrdered) AS total_units_sold
FROM products p INNER JOIN orderdetails as o
ON p.productCode = o.productCode
GROUP BY p.warehouseCode
ORDER BY total_units_sold DESC;
```

Query 7: **COPIED**

```
USE mintclassics;
SELECT p.warehouseCode,
SUM(p.quantityInStock) AS inventory_units,
SUM(o.quantityOrdered) AS units_sold,
SUM(o.quantityOrdered) / SUM(p.quantityInStock)
AS inventory_turnover
FROM products AS p INNER JOIN orderdetails AS
o ON p.productCode = o.productCode
GROUP BY p.warehouseCode
ORDER BY inventory_turnover;
```



Appendix

SQL queries cont.

Query 8:

```
USE mintclassics;
SELECT p.warehouseCode,
SUM(o.quantityOrdered * o.priceEach) AS revenue
FROM products AS p INNER JOIN orderdetails as
o
ON p.productCode = o.productCode
GROUP BY p.warehouseCode
ORDER BY revenue DESC;
```

Query 9:

```
USE mintclassics;
SELECT p.warehouseCode, p.productCode,
p.productName, p.quantityInStock,
SUM(o.quantityOrdered) AS total_sold
FROM products AS p LEFT JOIN orderdetails as o
ON p.productCode = o.productCode
GROUP BY p.warehouseCode, p.productCode,
p.productName, p.quantityInStock
ORDER BY total_sold ASC;
```